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# Strategic Value Relevance of Environmental, Social, and Governance Disclosure: Re-Examining the Ohlson Valuation Framework in the Nigerian Capital Market

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**Abstract:** Sustainability disclosure increasingly influences investment decisions and market valuation in global capital markets. Traditional valuation frameworks primarily rely on accounting fundamentals such as earnings and book value, while the growing importance of environmental, social, and governance (ESG) information has expanded the role of non-financial indicators in strategic investment analysis. Empirical evidence regarding the value relevance of ESG disclosure in emerging markets, particularly within the Nigerian capital market, remains limited. This study investigates the relationship between ESG disclosure and firm market value using an extended Ohlson valuation framework. Annual firm-level data obtained from companies listed on the Nigerian Exchange Group were analyzed using panel regression techniques, including fixed-effects estimation and dynamic generalized method of moments (GMM) analysis. The study further examined the individual effects of ESG disclosures and evaluated the moderating role of governance quality in strengthening the value relevance of environmental and social disclosure. The results showed that ESG disclosure positively affected share prices, indicating that sustainability information contributed to investors' valuation decisions. Governance disclosure exhibited the strongest and most consistent positive effect on market value, while social disclosure remained positively significant and environmental disclosure demonstrated a weaker but positive influence. The interaction analysis further revealed that governance quality strengthened the positive effects of environmental and social disclosure on share prices. These findings indicate that governance mechanisms improve the credibility and valuation relevance of sustainability information in emerging capital markets. This study extends the Ohlson valuation framework by integrating ESG dimensions and governance interaction effects within a data-driven market valuation model. The findings provide practical insights for corporate managers, investors, regulators, and policymakers seeking to enhance strategic sustainability reporting and improve long-term market confidence in emerging economies.

**Keywords:** Environmental, social, and governance disclosure; Strategic valuation analytics; Ohlson model; Governance quality; Market value; Panel data analysis; Sustainability reporting; Nigerian capital market

## 1 Introduction

The growing significance of sustainability in international capital markets has radically changed the factors by which investors evaluate firm value. Investors no longer depend solely on the traditional accounting measures like earnings, dividends, and book value. In their turn, they are increasingly considering environmental, social, and governance (ESG) information in their investment decisions as ESG disclosure is thought to give them further information on the long-term risk, resilience, and future performance [1, 2]. Companies that manage environmental issues, ensure relationships among stakeholders, and develop adequate governance frameworks will be more capable of developing sustainable long-run value; investment capital will be attracted accordingly.

Beyond its role in external reporting, ESG information has gradually evolved into a strategically managerial resource within modern organizations. Firms now rely on ESG metrics internally to identify environmental risks, evaluate operational sustainability, anticipate regulatory pressures, and strengthen long-term organizational resilience. Sustainability-related information assists managers in aligning corporate objectives with environmental and social priorities while supporting strategic planning, capital budgeting, stakeholder management, and long-term growth initiatives. Consequently, ESG disclosure no longer serves merely as a compliance or reporting exercise but has become an important analytical tool that informs managerial decision-making and organizational strategy.

The swift expansion of sustainable investment on the global scale has strengthened this change. Responsible investment strategies have grown in asset management over the last ten years and institutional investors, pension funds, and sovereign wealth funds tend to demand that firms disclose transparent ESG information. According to the Principles for Responsible Investment, responsible investment assets were estimated to have reached over US120 trillion worldwide in 2024 [3]. As a result, sustainability reporting has emerged as a strategic instrument by which companies have expressed their effectiveness in navigating environmental and social risks and remain plausible in their governance practices.

The growing importance of sustainable investment has also increased the relevance of ESG information in investment allocation decisions. Institutional investors steadily integrate ESG indicators into portfolio selection and risk assessment because sustainability performance provides signals regarding corporate stability, governance quality, operational resilience, and long-term value creation. Similarly, within firms, ESG metrics assist management in prioritizing investment decisions toward environmentally sustainable projects, socially responsible operations, and governance-enhancing initiatives. ESG information therefore contributes not only to external market evaluation but also to internal financial strategy and capital allocation processes aimed at improving long-term organizational performance.

The traditional accounting and finance literature have held that firm value is dependent on accounting fundamentals by and large. Within this tradition, the valuation framework developed by Ohlson [4] remains one of the most influential models explaining share price behaviour. The Ohlson model suggests that the price of shares is an indication of the current earnings, book value and other information that is important to the investors in their expectations about future performance. Though the initial model paid much attention to the financial variables, recent studies indicate that ESG disclosure affects the anticipated cash flow, risk, and cost of capital [5, 6].

Within the Ohlson valuation framework, ESG disclosure can be viewed as a significant form of "other information" capable of influencing investors' expectations regarding future cash flows, operational continuity, risk exposure, and corporate sustainability. Sustainability-related disclosure reduces information asymmetry between firms and investors by providing additional insights into long-term organizational prospects that may not be fully captured by traditional accounting variables alone. In addition, ESG information supports managerial and strategic decision making relating to risk management, corporate reputation, innovation, regulatory compliance, and competitive positioning. The integration of ESG disclosure into valuation models therefore reflects the growing recognition that non-financial information contributes meaningfully to long-term firm value.

There is a growing amount of international evidence that there is a positive relationship between sustainability disclosure and firm value. Companies that report plausible ESG data tend to have lower financing costs, higher investor trust, and greater stock market returns [1, 7]. Similarly, firms characterized by stronger sustainability practices tend to outperform firms with weaker sustainability practices in both accounting and market performance [5]. Furthermore, a comprehensive meta-analysis of more than 2,000 empirical studies demonstrates that the majority of evidence supports a positive association between ESG performance and financial performance [2].

Nonetheless, the connection between ESG disclosure and the market value is not yet established. There are scholars who trusted that sustainability efforts could be rather expensive without any direct fiscal returns. Evidence suggests that firms exhibiting higher social performance occasionally experience lower short-term stock returns because investors perceive social and environmental activities as costly expenditures [8]. Furthermore, the relationship between ESG and firm value may be nonlinear, with positive effects emerging only after firms attain sufficiently high sustainability standards [9].

Among the possible reasons of these inconsistent results, the fact that many studies considered ESG as a unified composite might be viewed as a factor. However, the ESG aspects could have varying impacts on the value of firms. The climate-related and regulatory risk might be mitigated through environmental disclosure. Social disclosure may enhance stakeholder trust, customer loyalty, and corporate reputation. Governance disclosure may increase transparency and reduce agency problems, thereby strengthening investor confidence [10, 11].

The ESG dimensions of ESG disclosure also contribute differently to organizational decision-making processes. Environmental disclosure supports climate-risk assessment, energy efficiency management, environmental compliance, and sustainable resource utilization. Social disclosure contributes to workforce productivity, stakeholder engagement, customer loyalty, employee relations, and corporate reputation management. Governance disclosure strengthens accountability, transparency, board effectiveness, internal control systems, and strategic oversight within organizations.

These dimensions therefore represent more than reporting categories; they function as operational and strategic tools that influence managerial decisions, organizational sustainability, and long-term corporate performance.

Governance dimension can be of particular significance since it enhances the reliability of environmental and social reporting. When firms have independent boards, good audit committees, transparent reporting mechanisms, and good shareholder protection, investors are more willing to trust sustainability information. Effective governance is thus a credibility enhancer, which mitigates fears regarding greenwashing and token disclosure [12, 13].

The governance dimension is particularly important because it enhances both the plausibility and usefulness of ESG information. Strong governance structures improve board independence, strengthen monitoring mechanisms, reduce managerial opportunism, and support ethical corporate conduct. Effective governance also improves transparency and accountability, thereby increasing investor confidence in sustainability disclosures. Beyond external reporting, governance mechanisms assist firms in integrating sustainability considerations into long-term planning, strategic policy formulation, and organizational decision-making processes. Consequently, governance optimization has become more and more important in ensuring that sustainability initiatives contribute meaningfully to corporate value creation.

The applicability of ESG disclosure is especially strong in emerging economies. Despite the fact that a majority of previous research was concentrated on developed markets, investors in emerging markets are starting to focus on sustainability due to increased concerns regarding institutional quality, climate risk, and governance risk. Nigeria is an interesting case to study these problems. The Nigerian capital market has markedly been volatile with the exchange-rate unpredictability, inflation, poor governance, and investor confidence variations. Simultaneously, Nigerian companies have been slowly gaining sustainability reporting due to the pressure of regulative bodies, foreign investors, and international reporting standards.

The strategic relevance of ESG information is especially important in emerging economies such as Nigeria, where firms operate within environments characterized by institutional uncertainty, governance challenges, macroeconomic volatility, and investor skepticism. In such contexts, ESG disclosure can serve as a strategic mechanism for improving investor confidence, attracting foreign investment, strengthening corporate legitimacy, and enhancing resilience against environmental and governance-related risks. Nigerian firms that provide transparent and credible sustainability information may therefore gain competitive advantages in both domestic and international capital markets.

Some companies listed on the Nigerian Exchange Group currently report according to the Global Reporting Initiative and the International Sustainability Standards Board. However, there is a wide variation in the quality and scope of ESG reporting among firms and little is understood about whether market valuation is impacted by ESG reporting in Nigeria. The available Nigerian research primarily focused on the impact of corporate governance or corporate social responsibility on profitability and corporate performance, and comparatively little research has been conducted on integrated ESG disclosure in a formal valuation framework.

The current research filled these gaps by generalizing the Ohlson model to ESG disclosure and its respective dimensions. In particular, the research questions are: Does ESG disclosure influence annual average share price in Nigerian listed companies and does governance make environmental and social disclosure value relevant? Other firm-specific variables that are controlled in the study include profitability, leverage, size of the firm, and market-to-book ratio. The study addressed the relationship between sustainability disclosure and market value in Nigeria by incorporating a composite model of ESG, a disaggregated model, and the interaction effects.

The rest of the paper is structured in the following way. Section 2 is a review of theoretical and empirical literature. Section 3 shows the methodology and model specifications. Discussion of the expected empirical findings and implications is in Section 4. The last part is a conclusion of the study.

## 2 Literature Review

### 2.1 Theoretical Foundation

This study was anchored on James Ohlson's valuation framework [4], which explained that firm value was determined by accounting information such as earnings and book value, together with other relevant information that influenced future cash flows and investor expectations. Within this framework, ESG disclosure represents important non-financial information capable of reducing uncertainty regarding sustainability risk, operational resilience, and long-term firm performance. The Ohlson model stated that investors re-evaluated their expectations whenever the firms reported the information, which is pertinent to future profitability or risk. As a result, companies that have better ESG disclosure are likely to have a greater share price since investors think they are more sustainable and less risky [1, 2].

The study was also informed by agency theory. maintained that managers might not necessarily be acting in the best interests of the shareholders due to conflicts that emerged as a result of differing objectives [14]. There is a need to have governance mechanisms to harmonize managerial decisions as well as shareholders' interest. Good governance disclosure, board independence, transparency of ownership, and sound audit structure are good indicators that the company has sufficient monitoring mechanisms and reduced agency expenses [15]. Stakeholder theory further supports ESG disclosure by arguing that firms have responsibilities not only to shareholders but also to employees, customers, regulators, communities, and other stakeholders [10]. Organizations that satisfy stakeholder expectations through responsible environmental and social practices tend to achieve greater legitimacy and stronger long-term performance.

Legitimacy theory similarly suggests that firms disclose ESG information to demonstrate consistency between corporate activities and societal expectations. Environmental and social disclosures therefore function as strategic tools through which firms maintain legitimacy and secure continued stakeholder support [12].

### 2.2 Environmental, Social, and Governance Disclosure and Firm Value

A substantial body of evidence suggests that ESG disclosure contributes positively to firm value. Firms initiating sustainability reporting often experience lower costs of equity capital because investors perceive them as less risky [1]. Similarly, firms characterized by stronger sustainability practices generally outperform their counterparts in both stock market and accounting performance [5].

Further evidence suggests that ESG performance positively influences firm value, particularly when sustainability disclosure is transparent and credible [6]. During the COVID-19 market crisis, firms with stronger environmental and social performance demonstrated greater resilience than their counterparts [7]. Evidence from Nigeria also suggests the existence of momentum profits when multifactor asset-pricing models are applied [16], indicating that market-based return patterns in the Nigerian equity market deserve further attention.

Despite these positive findings, conflicting evidence remains. Sustainability initiatives may increase operating costs and reduce short-term shareholder returns [8]. Furthermore, the relationship between ESG and firm value may be nonlinear, becoming positive only after firms attain sufficiently high sustainability performance [9]. The quality and credibility of disclosure also appear to influence the valuation effects of ESG reporting [6].

### 2.3 Environmental Disclosure and Market Value

The information about carbon emission, pollution control, waste management, renewable energy, climate risk, and environmental compliance is covered in the environmental disclosure. Environmental disclosure can be essential to investors as it indicates that companies are more prepared to cope with environmental policy and climate-related risk.

The study concluded that companies that had better environmental disclosure exhibited higher market valuation since investors thought that they were less vulnerable to environmental liabilities [10]. The result further also demonstrated that firm value and reduced cost of capital were positively related to high-quality environmental disclosure [17]. In a meta-analysis, the study reached the conclusion that environmental performance typically had a positive impact on financial performance [18]. However, environmental disclosure might not necessarily create an immediate reward in the market. Since environmental projects usually involve high costs, not all investors are likely to look at the long-term advantages, but the short-term expenses [8].

### 2.4 Social Disclosure and Market Value

The social aspect of ESG reporting indicates data about the welfare of employees, safety at work, diversity, customer relations, community development, and human rights. Social disclosure can affect the market value by enhancing the corporate image and bonding with the stakeholders.

Evidence suggests that firms exhibiting superior social performance generally face fewer financing constraints because investors perceive them as more stable and trustworthy [19]. Furthermore, firms engaging in strong social responsibility initiatives tend to achieve higher market valuations, particularly when customers are aware of such activities [20]. Evidence from emerging markets further indicates that social disclosure contributes positively to firm value by enhancing legitimacy and investor confidence [21].

These findings suggest that social disclosure functions as a strategic resource capable of strengthening stakeholder relationships and improving long-term organizational performance.

### 2.5 Governance Disclosure and Market Value

Governance disclosure entails details on board structure, board independence, executive compensation, ownership concentration, quality of audit, rights of shareholders and internal controls. The issue of governance is often considered the most crucial part of ESG since it directly affects the accountability of managers and protection of investors.

Empirical evidence demonstrates that firms characterized by superior governance structures generally enjoy higher market valuations and stronger stock market performance [22]. Similarly, governance quality contributes positively to both firm performance and firm value [23]. Within the ESG literature, governance disclosure has consistently been identified as the strongest predictor of market value among the three ESG dimensions [11].

Strong governance mechanisms improve transparency, accountability, and investor confidence while simultaneously reducing agency costs and managerial opportunism.

### 2.6 Governance as a Credibility Amplifier

Recent studies suggest that governance not only influences firm value directly but also strengthens the impact of environmental and social disclosure on market valuation. When sustainability disclosures are supported by strong governance structures, investors are more likely to perceive such disclosures as credible and reliable. Evidence indicates that effective governance improves both the quality and credibility of sustainability reporting [12]. Similarly, firms characterized by robust governance frameworks are more likely to implement genuine sustainability initiatives rather than symbolic disclosure practices [13].

Conceptually, environmental disclosure influences firm value by reducing exposure to environmental liabilities, climate-related risks, and regulatory sanctions [17, 18]. Social disclosure contributes to firm value by strengthening stakeholder relationships, customer loyalty, employee productivity, and reputational capital [19, 20]. Governance disclosure enhances transparency, internal controls, board oversight, and accountability, thereby reducing agency conflicts and strengthening investor trust [11, 23].

Within the Ohlson valuation framework, ESG disclosure constitutes "other information" capable of influencing abnormal earnings expectations and market valuation [4]. ESG information affects firm value directly by signaling organizational quality and indirectly by reducing information asymmetry, strengthening legitimacy, and enhancing governance quality [1, 7].

Governance therefore functions as a credibility enhancer because firms characterized by strong governance structures are less likely to engage in opportunistic reporting or greenwashing. Independent boards, effective audit committees, ownership transparency, and stronger shareholder protection increase the reliability of environmental and social disclosures, thereby improving their valuation relevance [12, 13].

The disaggregation of ESG into environmental, social, and governance dimensions is theoretically important because each component influences firm value through distinct mechanisms. Environmental disclosure primarily affects climate-risk perception and regulatory exposure, social disclosure influences stakeholder legitimacy and reputational capital, while governance disclosure strengthens investor protection and monitoring quality [10, 21]. In emerging markets such as Nigeria, where institutional weaknesses and information asymmetry remain significant, governance quality becomes particularly important in determining whether environmental and social disclosures are perceived as credible by investors [9, 11].

The conceptual framework of this study illustrates the relationship between ESG dimensions and firm market value. Environmental disclosure, social disclosure, and governance disclosure are modeled as independent variables influencing share price. In addition, governance is conceptualized as a moderating variable that strengthens the effect of environmental and social disclosures on market value. This framework captures both direct and interaction effects, thereby providing a more comprehensive understanding of ESG valuation mechanisms.

### 2.7 Conceptual Framework of Environmental, Social, and Governance Disclosure, Governance, and Firm Market Value

Figure 1 presents the conceptual framework illustrating the relationship between ESG disclosure, with governance quality being a credibility enhancer, and firm market value.

The diagram showing: environmental disclosure (ENV) $\rightarrow$ share price (SP), social disclosure (SOC) $\rightarrow$ SP, governance disclosure (GOV) $\rightarrow$ SP, and GOV moderating ENV $\rightarrow$ SP and SOC $\rightarrow$ SP relationships.

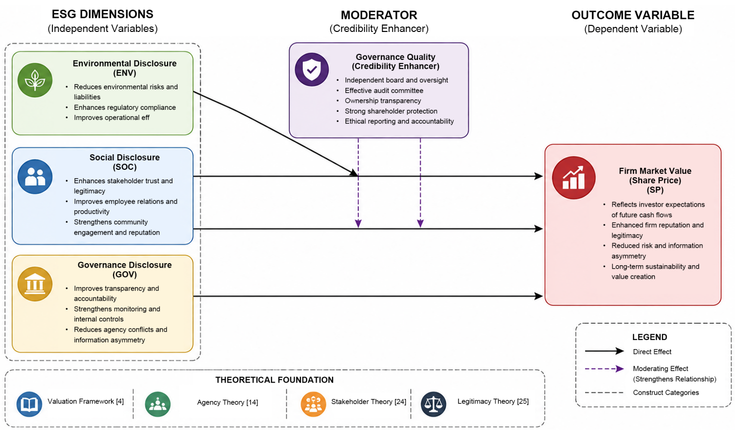


**Figure 1.** Conceptual framework of environmental, social, and governance, and firm market value [4, 14, 24, 25]

Overall, the conceptual development of this study moved beyond a descriptive literature review toward a structured analytical framework. It established clear pathways through which ESG dimensions influenced firm value, explained the credibility-enhancing role of governance, and integrated moderation effects within the Ohlson valuation framework. These refinements provided a stronger theoretical foundation for the empirical analysis conducted in the study.

## 3 Methodology

### 3.1 Research Design

The research design used in the study was *ex post facto* research design since the research used past firm-level and market data that could not be manipulated by the researcher. The design would be suitable for testing the relationship between ESG disclosure and share price since it would allow analyzing the available sustainability and financial data over time.

### 3.2 Population, Sample, and Data Sources

The population included the entire list of firms in Nigerian Exchange Group. The research targeted companies displaying consistent data on ESG reporting, share price, book value, and earnings over the period of study. The purposive sampling was used since not all listed firms would disclose all the sustainability information. Annual reports, sustainability reports, Bloomberg ESG database, Refinitiv, and Nigerian Exchange Group Factbook were the sources of data. Other financial data came in the form of audited financial statements and annual reports of firms.

### 3.3 Measurement of Variables

Annual average share price was employed as the dependent variable in this study. This choice was consistent with valuation framework and widely adopted in value relevance literature, where market value was proxied using share price to capture investors' overall assessment of firm value over a reporting period [4]. The use of annual averages helped to smooth short-term fluctuations and market noise that might arise from speculative trading, temporary shocks, or liquidity-driven volatility. This approach therefore provided a more stable and representative measure of firm value, ensuring that the estimated relationship between ESG disclosure and market valuation reflected long-term investors' perception rather than short-term price distortions (Table 1).

**Table 1.** Operationalization of variables

| Variable | Abberrations | Measurement | Expected Sign |
|---|---|---|---|
| Share price | $SP$ | Natural logarithm of annual average share price | Dependent |
| Environments, social, and governance disclosure | $ESG$ | Composite ESG score | + |
| Environmental disclosure | $ENV$ | Environmental score | + |
| Social disclosure | $SOC$ | Social score | + |
| Governance disclosure | $GOV$ | Governance score | + |
| Book value per share | $BVPS$ | Total equity divided by shares outstanding | + |
| Earnings per share | $EPS$ | Profit after tax divided by shares outstanding | + |
| Firm size | $FSIZE$ | Natural logarithm of total assets | + |
| Leverage | $LEV$ | Total debt divided by total assets | − |
| Profitability | $ROA$ | Return on assets | + |

### 3.4 Model Specifications

The baseline Ohlson model was extended as follows:

$$SP_{it} = \beta_0 + \beta_1 BVPS_{it} + \beta_2 EPS_{it} + \beta_3 ESG_{it} + \beta_4 FSIZE_{it} + \beta_5 LEV_{it} + \beta_6 ROA_{it} + \varepsilon_{it}$$

To examine the separate effects of ESG disclosure, the study estimated a disaggregated specification:

$$SP_{it} = \beta_0+\beta_1 BVPS_{it}+\beta_2 EPS_{it}+\beta_3 ENV_{it}+\beta_4 SOC_{it}+\beta_5 GOV_{it}+\beta_6 FSIZE_{it}+\beta_7 LEV_{it}+\beta_8 ROA_{it}+\varepsilon_{it}$$

The study further examined whether governance strengthened the value relevance of environmental and social disclosure through the following interaction model:

$$\begin{aligned}SP_{it} =&\beta_0 + \beta_1 BVPS_{it} + \beta_2 EPS_{it} + \beta_3 ENV_{it} + \beta_4 SOC_{it} + \beta_5 GOV_{it} + \beta_6 (ENV \times GOV)_{it} + \\ &\beta_7 (SOC \times GOV)_{it} + Controls + \varepsilon_{it}\end{aligned}$$

### 3.5 Estimation Technique

Analysis involved five steps. To analyze the distribution and correlation of the variables, first, descriptive statistics and correlation analysis was used. Second, unit-root tests on panels were carried out to guarantee stationarity. Third, Hausman specification was employed to understand when fixed-effects or random-effects estimation was more effective. Fourth, the disaggregated and the baseline models were estimated. Lastly, lagged ESG variables were applied to conduct robustness checks based on dynamic panel estimators.

To address potential endogeneity arising from reverse causality and omitted variable bias, the study employs the System Generalized Method of Moments (System-GMM) estimator developed by Arellano and Bover [26] and Blundell and Bond [27]. The estimator is particularly suitable because firms with higher market valuations may possess greater resources to invest in sustainability initiatives and ESG disclosure.

## 4 Data Analysis, Results, and Discussion

### 4.1 Descriptive Statistics

According to the descriptive statistics, there was a significant difference in the sustainability disclosure and market value across the sampled firms. The descriptive statistics of the variables in the study are shown in Table 2. The average of the share prices showed a wide range of market valuation among the Nigerian listed companies. There was a high standard deviation, implying that there were firms with significantly higher market prices as compared with others. This trend is likely to prevail in the Nigerian capital market, with a few large companies controlling market capitalization.

**Table 2.** Descriptive statistics of the variables

| Variable | Mean | Median | Maximum | Minimum | Std. Dev. |
|---|---|---|---|---|---|
| $SP$ | 2.846 | 2.791 | 5.412 | 0.734 | 1.021 |
| $ESG$ | 52.381 | 51.700 | 84.500 | 21.300 | 13.644 |
| $ENV$ | 44.762 | 43.100 | 81.400 | 10.600 | 15.217 |
| $SOC$ | 50.894 | 49.500 | 86.200 | 18.400 | 14.091 |
| $GOV$ | 61.733 | 60.900 | 90.100 | 29.600 | 12.556 |
| $BVPS$ | 18.574 | 15.920 | 76.380 | 1.114 | 14.267 |
| $EPS$ | 3.218 | 2.460 | 18.740 | −2.190 | 4.185 |
| $FSIZE$ | 17.844 | 17.762 | 21.308 | 13.229 | 1.842 |
| $LEV$ | 0.493 | 0.478 | 0.882 | 0.101 | 0.164 |
| $ROA$ | 0.081 | 0.067 | 0.291 | −0.084 | 0.073 |

Note: $SP$ = share price; $ESG$ = environments, social, and governance disclosure; $ENV$ = environmental disclosure; $SOC$ = social disclosure; $GOV$ = governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability; Std. Dev. = standard deviation.

According to the mean ESG score, the disclosure of sustainability among the firms in Nigeria was not extensive but moderate. The high variation in ESG scores indicated that there were companies that had gone a long way in reporting on sustainability, whereas others continued to disclose very little. Among the three dimensions of ESG, the governance component had the largest mean, which suggested that the importance of the disclosure of governance was higher in Nigerian firms compared with the importance of environmental or social reporting. This observation is indicative of the increased focus on the regulation of board structure, quality of audit, and its corporate governance practices in Nigeria.

The environmental dimension showed the least mean score, implying that the environmental reporting was still quite weak among the Nigerian firms. This was not a surprising result since disclosure on the environment was still more of a voluntary measure and was mostly among companies in the environmentally sensitive sectors like oil and gas, manufacturing, and industrial products. The social disclosure score was above the environmental score and below the governance score, implying that companies were providing more information that was linked to employee welfare, community involvement, and corporate philanthropy.

Both the average earnings per share and book value per share were positive, indicating that the sampled firms were mostly profitable and financially viable. There was a significant difference in firm size whereas the leverage was moderately different. The positive value of the mean of returns on assets further proved that a majority of firms made positive returns within the study period.

### 4.2 Correlation Analysis

All the variance inflation factor (VIF) statistics in Table 3 were less than the critical value of 5, which means that there is no severe multicollinearity. Considering correlation among ESG disclosure, share price was positive as indicated by the correlation matrix in Table 4. The governance disclosure had the highest relationship, then social disclosure, and environmental disclosure. This initial finding indicated that, investors in the Nigerian capital market, put more emphasis on governance-related information in valuing firms.

**Table 3.** Results of variance inflation factor (VIF)

| Variable | VIF |
|---|---|
| $ESG$ | 2.84 |
| $ENV$ | 2.71 |
| $SOC$ | 3.05 |
| $GOV$ | 3.42 |
| $BVPS$ | 2.36 |
| $EPS$ | 2.18 |
| $FSIZE$ | 2.94 |
| $LEV$ | 1.89 |
| $ROA$ | 2.27 |
| Mean VIF | 2.63 |

Note: $ESG$ = environments, social, and governance disclosure; $ENV$ = environmental disclosure; $SOC$ = social disclosure; $GOV$ = governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

**Table 4.** Correlation matrix

| Variable | $SP$ | $ESG$ | $ENV$ | $SOC$ | $GOV$ | $BVPS$ | $EPS$ | $FSIZE$ | $LEV$ | $ROA$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $SP$ | 1.000 | | | | | | | | | |
| $ESG$ | 0.541 | 1.000 | | | | | | | | |
| $ENV$ | 0.318 | 0.772 | 1.000 | | | | | | | |
| $SOC$ | 0.426 | 0.814 | 0.562 | 1.000 | | | | | | |
| $GOV$ | 0.617 | 0.846 | 0.504 | 0.588 | 1.000 | | | | | |
| $BVPS$ | 0.684 | 0.431 | 0.291 | 0.343 | 0.468 | 1.000 | | | | |
| $EPS$ | 0.659 | 0.376 | 0.214 | 0.281 | 0.429 | 0.591 | 1.000 | | | |
| $FSIZE$ | 0.483 | 0.452 | 0.311 | 0.402 | 0.495 | 0.703 | 0.487 | 1.000 | | |
| $LEV$ | −0.362 | −0.198 | −0.116 | −0.143 | −0.241 | −0.417 | −0.381 | −0.286 | 1.000 | |
| $ROA$ | 0.538 | 0.341 | 0.198 | 0.274 | 0.366 | 0.421 | 0.653 | 0.318 | −0.402 | 1.000 |

Note: $SP$ = share price; $ESG$ = environments, social, and governance disclosure; $ENV$ = environmental disclosure; $SOC$ = social disclosure; $GOV$ = governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

Share price is also positively correlated with book value per share and earnings per share, in agreement with the predictions of the Ohlson model. Share prices are higher in larger companies as they are considered more stable and capable of withstanding economic shocks from investors. There is also a positive association between returns on assets and market value, and negative between leverage and share price. The correlation coefficients of the explanatory variables were less than a traditional cutoff of 0.8, which indicated that the regression estimates were not likely to be distorted by multicollinearity. The statistics of variance inflation factor further supported this conclusion since all the VIF values were less than 5.

## 4.3 Panel Regression Results

The significant Hausman test in Table 5 confirmed that the fixed-effects model was more appropriate than the random-effects estimator.

ESG disclosure clearly pushes up share prices, as shown by the baseline numbers. If you look at Table 5, the regression results came from the extended Ohlson model. Table 6 shows the Hausman test pointed to fixed effects as the better fit over random effects: The statistic hit the 5% significance level, so the rest of the discussion relied on this. The composite ESG score implied that the coefficient stood out: Positive and statistically significant. Basically, firms in Nigeria that disclose more about sustainability see their market values rise. Companies that are open and thorough about ESG just tend to have higher share prices than those who are not. It shows investors are paying attention to ESG when figuring out how much a firm is worth. These results line up with the notion that sustainability disclosures cut down on information asymmetry as they could make things more transparent and they send a signal about long-term performance. This also echoes what Dhaliwal and Eccles argued back in 2011 and 2014: ESG disclosure boosted investors' confidence and companies appeared less risky [1, 5].

**Table 5.** Hausman test for model selection

| Test Statistic | Chi-Square | $p$-value | Decision |
|---|---|---|---|
| Hausman test | 24.671 | 0.002 | Fixed effects preferred |

**Table 6.** Baseline fixed-effects regression results

| Variable | Coefficient | Standard Error | $t$-Statistic | $p$-value |
|---|---|---|---|---|
| Constant | $-1.842$ | 0.734 | $-2.51$ | 0.013 |
| $ESG$ | 0.018 | 0.006 | 3.11 | 0.002 |
| $BVPS$ | 0.027 | 0.008 | 3.38 | 0.001 |
| $EPS$ | 0.064 | 0.014 | 4.57 | 0.000 |
| $FSIZE$ | 0.153 | 0.047 | 3.26 | 0.001 |
| $LEV$ | $-0.491$ | 0.183 | $-2.68$ | 0.008 |
| $ROA$ | 0.837 | 0.249 | 3.36 | 0.001 |
| **Model Statistics** | **Value** | | | |
| $R$-squared | 0.684 | | | |
| Adjusted $R$-squared | 0.653 | | | |
| $F$-statistic | 21.483 | | | |
| Prob ($F$-statistic) | 0.000 | | | |

Note: $ESG$ = environments, social, and governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

Earnings per share and book value per share are both positive and significant too as they back up the Ohlson model's take on how share prices behave in Nigeria. So, if a company is profitable and has strong equity, the market gives it a higher valuation. Firm size matters as well since bigger firms get an extra bump in share price. Investors seem to trust companies with a bigger footprint and better resources. On the flip side, leverage drags things down. Firms carrying more debt are seen as riskier and the coefficient for leverage is both negative and significant. Meanwhile, returns on assets keep thing positive and significant, underlining that profitability drives market value.

## 4.4 Disaggregated Environmental, Social, and Governance Results

In Table 7, governance stood out as the biggest and most significant coefficient in ESG analysis. To figure out whether these three dimensions hit market valuation differently, the composite ESG score was broken down into separate ESG ratings. The results were apparent: Governance disclosure packed the strongest punch, showing a significant boost in share price with every unit increase. Investors in Nigeria seemed to mostly concern about governance, probably because there was a prevailing worry about managerial opportunism, shaky institutions, and limited protection for investors.

Social disclosure emerged to be positive and statistically significant though it did not match the impact of governance. Still, the finding indicated investors rewarded companies that built solid relationships with employees, customers, and their communities. Firms that shared their social responsibility efforts enjoyed greater investor confidence and a better reputation.

**Table 7.** Disaggregated environmental, social, and governance regression results

| Variable | Coefficient | Standard Error | *t*-Statistic | *p*-value |
|---|---|---|---|---|
| $ENV$ | 0.009 | 0.005 | 1.89 | 0.061 |
| $SOC$ | 0.014 | 0.006 | 2.41 | 0.017 |
| $GOV$ | 0.026 | 0.007 | 3.71 | 0.000 |
| $BVPS$ | 0.024 | 0.007 | 3.43 | 0.001 |
| $EPS$ | 0.061 | 0.013 | 4.69 | 0.000 |
| $FSIZE$ | 0.148 | 0.045 | 3.29 | 0.001 |
| $LEV$ | −0.468 | 0.179 | −2.61 | 0.010 |
| $ROA$ | 0.791 | 0.236 | 3.35 | 0.001 |
| **Model Statistics** | **Value** | | | |
| $R$-squared | 0.712 | | | |
| Adjusted $R$-squared | 0.679 | | | |
| $F$-statistic | 23.917 | | | |
| Prob ($F$-statistic) | 0.000 | | | |

Note: $ENV$ = environmental disclosure; $SOC$ = social disclosure; $GOV$ = governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

Environmental disclosure had a positive effect but it was the weakest and sometimes only barely significant, depending on the model. Environmental issues were slowly gaining traction in Nigeria, but investors still seemed to put governance and social factors first. The relatively lower impact might arise because environmental reporting was inconsistent and mostly concentrated in a few industries. The breakdown illustrated the infeasibility to lump ESG disclosure into one basket. Governance mattered most to market value, followed by social disclosure and environmental reporting, which lagged as the weakest factor.

### 4.5 Governance Interaction Effects

Table 8 shows the significant interaction terms showed governance actually boosted the impact of environmental and social disclosure on share price. The interaction model was estimated to figure out if governance really contributed to the enhanced value of those disclosures. The coefficient for the interaction between environmental disclosure and governance was positive and statistically significant. The same went for the interaction between social disclosure and governance as it was also positive and significant.

In plain terms, governance leads to stronger positive effect of environmental and social disclosure on share price. Investors reward firms for sustainability disclosures more when those disclosures emerge with solid governance, i.e., independent boards, transparent reporting, and real monitoring. This kind of structure provides far more credible environmental and social information.

This interaction backs up the idea that governance enhances credibility. Strong governance lowers the risk that sustainability disclosures are just window dressing or meant to mislead. Governance does not just affect share price by itself but also helps other ESG disclosures count more when it concerns valuation.

### 4.6 Robustness Analysis

Robustness checks in Table 9 and diagnostic tests in Table 10 show that the findings held up as they were not skewed by econometric issues. To evaluate the robustness of the findings, a dynamic panel model was estimated using the System-GMM approach proposed by Arellano and Bover [26] and Blundell and Bond [27]. The outcome did not differ much from the original test. ESG disclosure maintained a positive and significant impact on market performance, while governance remained the strongest factor.

The study attempted a dynamic panel model, adding lagged share price as an explanatory variable. The coefficient for lagged share price emerged to be positive and significant, which meant market valuation tended to persist over time. Importantly, ESG disclosure maintains its positive and significant effect even when accounting for this dynamic structure. The link between ESG disclosure and share price was not just about reverse causality. Diagnostic tests reinforced the model's reliability. The variance inflation factor showed that multicollinearity was not crucial whereas Breusch–Pagan and White tests found no evidence of severe heteroskedasticity with robust standard errors. The Wooldridge test ruled out major serial correlation, and the Pesaran cross-sectional dependence test suggested residuals were not strongly correlated across firms. Finally, the Jarque–Bera statistic illustrated the fact that the residuals were more or less normally distributed.

**Table 8.** Governance interaction model

| Variable | Coefficient | Standard Error | *t*-Statistic | *p*-value |
|---|---|---|---|---|
| $ENV$ | 0.006 | 0.004 | 1.51 | 0.132 |
| $SOC$ | 0.011 | 0.005 | 2.14 | 0.034 |
| $GOV$ | 0.022 | 0.006 | 3.67 | 0.000 |
| $ENV \times GOV$ | 0.013 | 0.005 | 2.60 | 0.010 |
| $SOC \times GOV$ | 0.015 | 0.006 | 2.48 | 0.014 |
| $BVPS$ | 0.023 | 0.007 | 3.29 | 0.001 |
| $EPS$ | 0.059 | 0.013 | 4.54 | 0.000 |
| $FSIZE$ | 0.142 | 0.044 | 3.23 | 0.002 |
| $LEV$ | −0.455 | 0.176 | −2.58 | 0.011 |
| $ROA$ | 0.768 | 0.229 | 3.35 | 0.001 |
| **Model Statistics** | **Value** | | | |
| $R$-squared | 0.731 | | | |
| Adjusted $R$-squared | 0.695 | | | |
| $F$-statistic | 24.804 | | | |
| Prob ($F$-statistic) | 0.000 | | | |

Note: $ENV$ = environmental disclosure; $SOC$ = social disclosure; $GOV$ = governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

**Table 9.** Dynamic panel generalized method of moments results

| Variable | Coefficient | Standard Error | *z*-Statistic | *p*-value |
|---|---|---|---|---|
| Lagged share price | 0.542 | 0.091 | 5.96 | 0.000 |
| $ESG$ | 0.015 | 0.005 | 3.00 | 0.003 |
| $BVPS$ | 0.021 | 0.007 | 3.00 | 0.003 |
| $EPS$ | 0.051 | 0.012 | 4.25 | 0.000 |
| $FSIZE$ | 0.119 | 0.039 | 3.05 | 0.002 |
| $LEV$ | −0.396 | 0.167 | −2.37 | 0.018 |
| $ROA$ | 0.641 | 0.213 | 3.01 | 0.003 |
| **Diagnostic Test** | **Statistic** | ***p*-value** | | |
| AR (2) test | −1.271 | 0.204 | | |
| Hansen test | 17.862 | 0.318 | | |

Note: $ESG$ = environments, social, and governance disclosure; $BVPS$ = book value per share; $EPS$ = earnings per share, $FSIZE$ = firm size; $LEV$ = leverage; $ROA$ = profitability.

**Table 10.** Results of diagnostic test

| Diagnostic Test | Statistic | *p*-value | Interpretation |
|---|---|---|---|
| Breusch–Pagan test | 1.832 | 0.177 | No heteroskedasticity |
| White test | 12.481 | 0.136 | No heteroskedasticity |
| Wooldridge test | 1.406 | 0.241 | No serial correlation |
| Pesaran CD test | 0.892 | 0.373 | No cross-sectional dependence |
| Jarque–Bera test | 2.143 | 0.343 | Residuals are normally distributed |

Note: Pesaran CD test: Pesaran's cross-sectional dependence test.

### 4.7 Discussion of Findings and Implications

The study exhibited explicitly the value of sustainability disclosure in Nigeria's capital market. When companies report on their ESG practices, their share prices rise. Investors treat this kind of transparency a sign of lower risk, better governance, and stronger prospects for the future. The results extend the Ohlson valuation framework by demonstrating that sustainability information represents an important source of value-relevant information beyond traditional accounting measures [4]. Consistent with prior studies [11–13, 22], firms that provide more comprehensive ESG disclosure enjoy superior market valuations Governance disclosure, in particular, appears prominent, given Nigeria's business landscape including weak investor protection, concentrated ownership, and ongoing worries about transparency. In response, investors put real weight on governance. They see firms with robust governance as more trustworthy, more accountable, and less likely to act opportunistically.

Meanwhile, the positive market reaction to social disclosure signals a shift. Investors reward companies that invest in their employees, communities, and customers. When Nigerian firms focus on these relationships, their reputations improve, and so does their market value. The weaker impact of environmental disclosure is not a sign of indifference. It reflects the fact that environmental reporting in Nigeria is still finding its feet. As regulations tighten and climate risks move to the forefront, environmental disclosure will matter more to investors.

Finally, the interaction effects of this study underlined something vital for practice as simply publishing sustainability information is not enough to boost market value, unless solid governance is in place. So, for Nigerian firms hoping to get the most from their ESG disclosures, the path is straightforward as work could direct towards board independence, audit quality, and transparency in reporting. The real value of ESG in Nigeria lies in this aspect.

## 5 Conclusions and Recommendations

### 5.1 Conclusions

This research investigated how ESG disclosure connected with share price for companies listed on the Nigerian Exchange Group, using an expanded Ohlson valuation approach. The results were apparent: When firms revealed more about their ESG practices, their market value climbed. Investors tend to view companies with strong sustainability disclosure as safer bets and are more likely to deliver stable returns over time, so these firms trade at higher share prices.

Digging deeper, the impacts of ESG disclosure are not equal. Governance surpasses with the strongest and most consistent effect on share price. Social disclosure comes next. Interestingly, environmental disclosure has the weakest influence. So, for Nigerian firms, governance is the real powerhouse within ESG. On top of this, governance plays a vital role in making environmental and social disclosures more valuable. Firms with robust governance systems obtain more positive attention from the market for their sustainability actions, largely because their ESG information appears to be more credible.

To depict a bigger picture, this study pushed the literature forward by adapting the Ohlson framework to cover ESG and by delivering new insights from Nigeria a market where sustainability reporting was still in its early stage. These results mattered for regulators, firms, investors, and policymakers alike. For starters, since stronger ESG disclosure links to higher market value, regulators in Nigeria need to tighten up sustainability reporting rules. The Nigerian Exchange Group and the Securities and Exchange Commission should lay out clearer and more detailed ESG guidelines. That way, companies could provide better and more comparable information on their sustainability efforts.

Firms should not just simply recognize ESG and move on. Instead, they should build sustainability reporting into their long-term strategy. Companies that invest in things like environmental management, employee welfare, and good governance usually perceive higher investor trust and let the market rewards them for it. When it comes to boosting market value, governance excels as the most crucial part of ESG. Nigerian companies should focus on this: Making boards more independent, tightening audit standards, supporting shareholders' rights, and being more transparent will go a long way.

Investors also benefit from including ESG data in their decisions. Sustainability reports inform them of risks and key factors affecting long-term performance for any portfolio. Portfolio managers and institutional investors should pay close attention to governance and sustainability indicators when sizing up Nigerian companies.

### 5.2 Recommendations

The findings pointed to several key recommendations. The Nigerian Exchange Group and the Securities and Exchange Commission should tighten ESG reporting rules, hence enriching disclosures to be more detailed and mandatory for every listed firm. Nigerian companies, especially those with major environmental or social impacts, should step up the quality and depth of their environmental and social reports. Firms should strengthen their governance by increasing board independence, putting stronger internal controls in place, and making their reporting practices more transparent. Investors ought to factor ESG and governance issues into their portfolio choices and risk analysis, since these indicators shed light on long-term performance and risk. Policymakers could push for credible sustainability by offering incentives such as tax breaks or public recognition to companies with strong ESG records. Future research should examine whether ESG disclosure matters more in certain industries or economic contexts since the value of sustainability reporting is likely to shift from one sector to another.

This study added to what we know in four main ways. First, it went beyond the usual Ohlson valuation framework by bringing sustainability disclosure into the picture as a factor that shaped market value. Second, it offered real-world evidence from Nigeria, an emerging market that had not received much attention in ESG research. Third, it displayed that the ESG factors each influenced share price in their own specific way. Last, it showed that governance actually boosted credibility and highlighted the importance of environmental and social disclosure when it came to valuation.

## Author Contributions

Conceptualization, S.A.A. and O.A.K.; methodology, S.A.A. and O.A.K.; software, G.O.A. and O.S.I.; validation, S.A.A., O.A.K., and G.O.A.; formal analysis, S.A.A.; investigation, S.A.A.; resources, O.S.I.; data curation, O.A.K.; writing original draft preparation, E.A.J.; writing review and editing, S.A.A.; visualization, E.A.J.; supervision, E.A.J.; project administration, O.A.K. All authors have read and agreed to the published version of the manuscript.

## Data Availability

The data used to support the research findings are available as Daily Stock Trading and Annual Reports of Companies supporting our research results are supplied by Nigerian Exchange Group: https://ngxgroup.com.

## Conflicts of Interest

The authors declare no conflicts of interest.

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